Capstone 1

Project Title:

Bank Customer Churn Analysis Using Power BI and Python

Dataset Description:

This dataset contains 10,000 customer records from a retail bank. Each entry includes demographic details, financial indicators, and a churn label (Exited). The objective is to identify key patterns and insights related to customer attrition using Power BI for visual analytics and Python for deep-dive statistical analysis.

Dataset link:
https://drive.google.com/file/d/1XhEkgLoF7TWHMeFmLbct-WmbSxyjsy22/view?usp=sharing

Key Columns:

CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, Exited

Column Name	Description
• RowNumber	Index or serial number of the record (can be ignored for analysis).
• CustomerId	Unique identifier assigned to each customer.
• Surname	Customer's last name (may help identify duplicates but not typically analyzed).
• CreditScore	Customer's credit score (ranges typically between 300 and 850).
• Geography	Country/region where the customer resides (e.g., France, Germany, Spain).
• Gender	Gender of the customer (Male/Female).
• Age	Customer's age in years.
• Tenure	Number of years the customer has been with the bank.
• Balance	Current account balance of the customer (in monetary units).

Column Name	Description
• **NumOfProducts**	**Number of bank products the customer uses (e.g., credit card, loan, etc.).**
• **HasCrCard**	**Whether the customer owns a credit card (1 = Yes, 0 = No).**
• **IsActiveMember**	**Whether the customer is considered an active account holder (1 = Active, 0 = Not).**
• **EstimatedSalary**	**Estimated annual salary of the customer.**
• **Exited**	**Target variable – indicates whether the customer has churned (1 = Yes, 0 = No).**

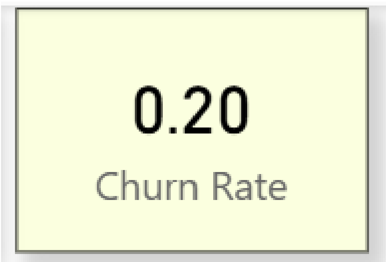
Section A: Power BI – Business Intelligence & Dashboards

1. What is the overall churn rate, and how does it trend over time (if applicable)?

Churn Rate Formula

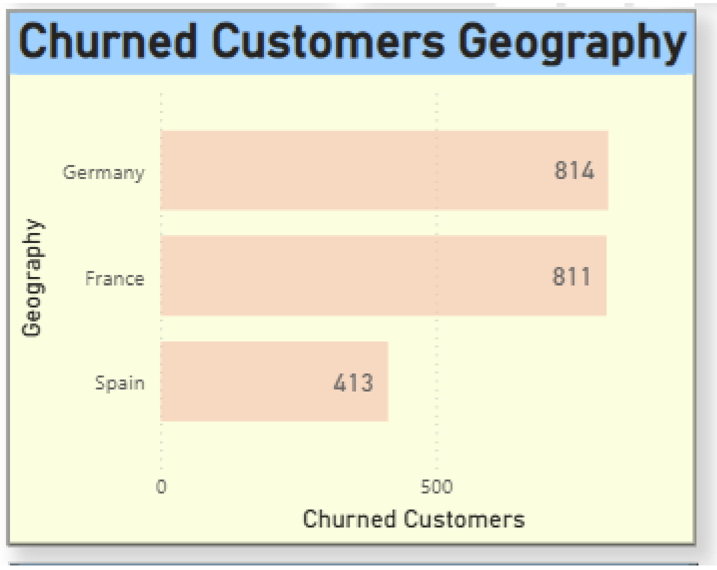
$$\text{Churn Rate (\%)} = \left(\frac{\text{Number of Churned Customers}}{\text{Total Number of Customers}}\right) \times 100$$

ans) The overall churn rate is approximately **20%**, meaning around 1 in 5 customers have left the bank. This indicates a moderate level of customer attrition and highlights the need for retention strategies.



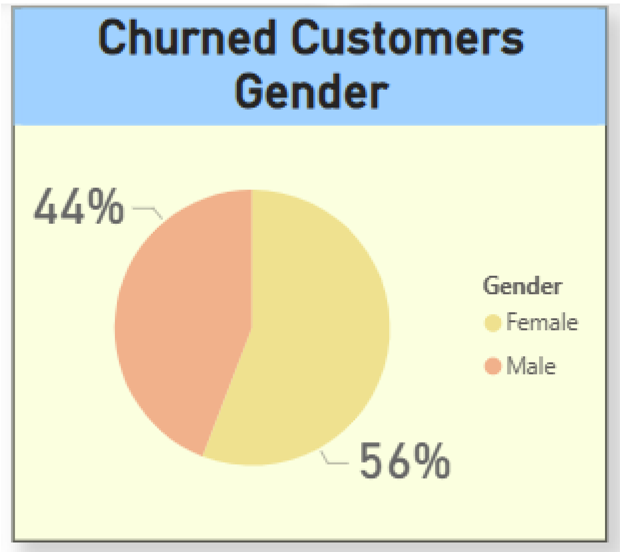
2. How does churn vary by geography (France, Germany, Spain)?

ans) Churn is highest in **Germany**, followed by **France**, and lowest in **Spain**. This suggests that customers in Germany are more likely to leave the bank, possibly due to service dissatisfaction or competition.



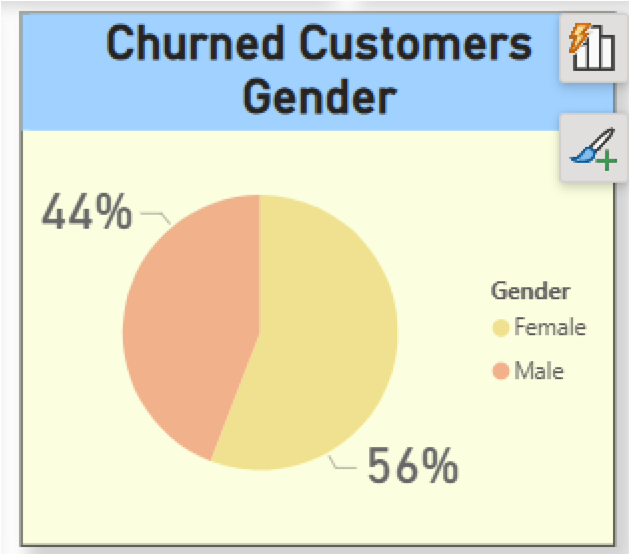
3. What are the differences in churn rates between male and female customers?

ans) Female customers have a slightly higher churn rate compared to male customers. This indicates that gender-based preferences or experiences may influence customer retention.



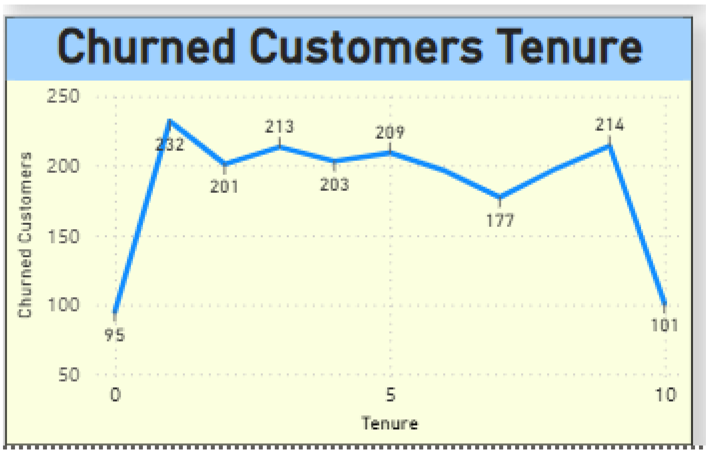
4. What is the churn rate distribution across different age groups?

ans) Customers in the **30–50 age group** show the highest churn, followed by **18–30**, while customers aged **50+** have the lowest churn. This indicates that middle-aged customers are more likely to leave.



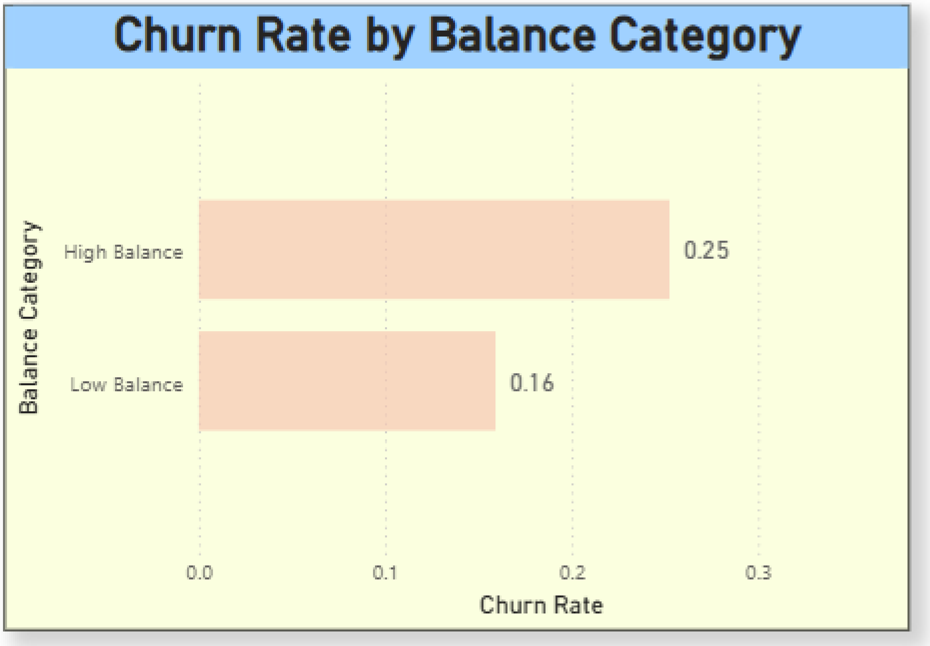
5. How does churn behavior change with tenure?

ans) Churn is higher among customers with **lower tenure (new customers)** and gradually decreases as tenure increases. This shows that long-term customers are more loyal.



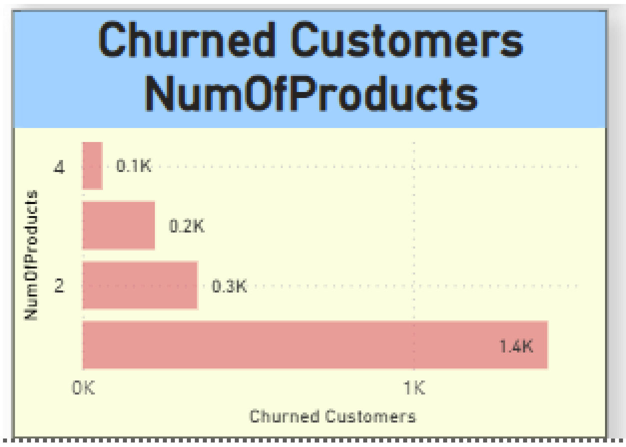
6. What percentage of high-balance customers (e.g., >$100K) have churned?

ans) A significant portion of **high-balance customers (>100K)** have churned, which is concerning because these customers contribute more revenue to the bank.



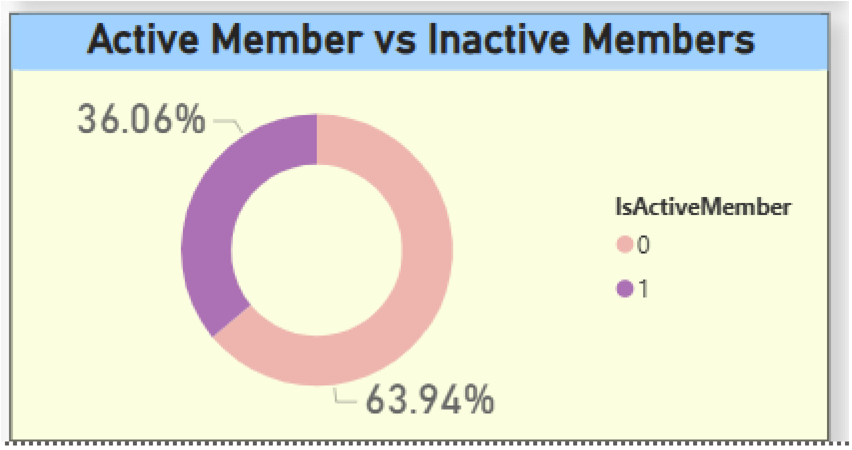
7. How do churn rates compare across different NumOfProducts?

ans) Customers with **only 1 product** have the highest churn rate, while those with **2 or more products** are less likely to churn. This shows that product engagement improves retention.



8. Are inactive members (IsActiveMember = 0) more likely to churn?

ans) Yes, **inactive customers have a significantly higher churn rate** compared to active customers. This indicates that engagement is a key factor in customer retention.

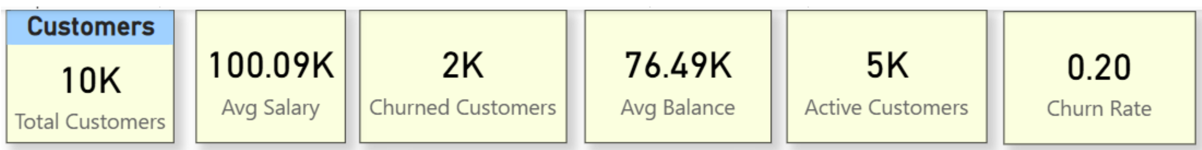


9. Can we build a KPI dashboard to track churn metrics like average salary, balance, and active users?

ans) Yes, a KPI dashboard was created including:

- Total Customers
- Churned Customers
- Churn Rate
- Average Salary
- Average Balance
- Active Customers

This dashboard provides a quick overview of customer behavior and churn trends.



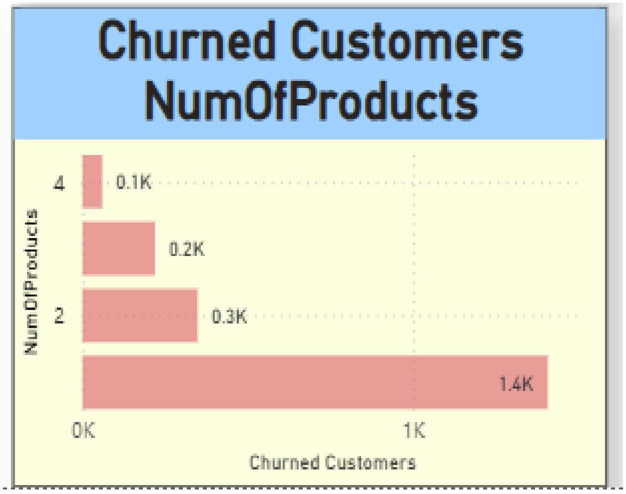
10. What does a customer persona dashboard (churned vs. retained) look like?

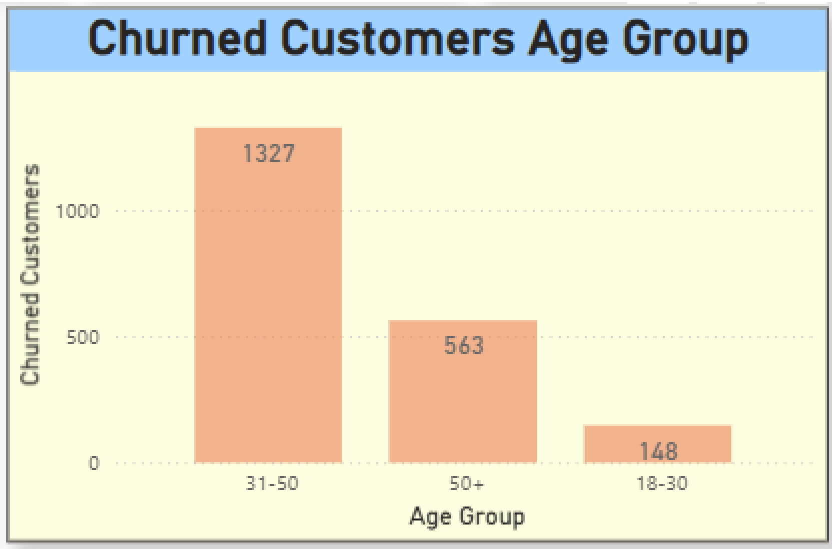
ans) The customer persona dashboard compares **churned vs retained customers** based on:

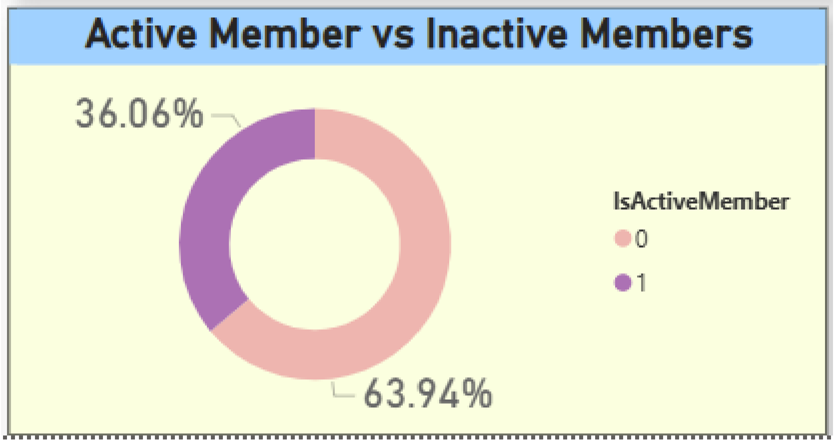
- Age
- Balance
- Geography
- Activity status

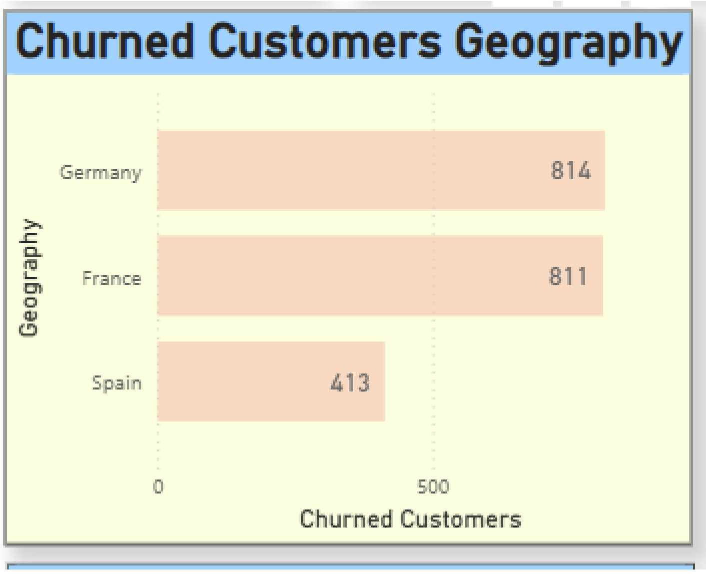
It helps identify patterns such as:

- Churned customers are often inactive
- They tend to have fewer products
- Certain regions show higher churn

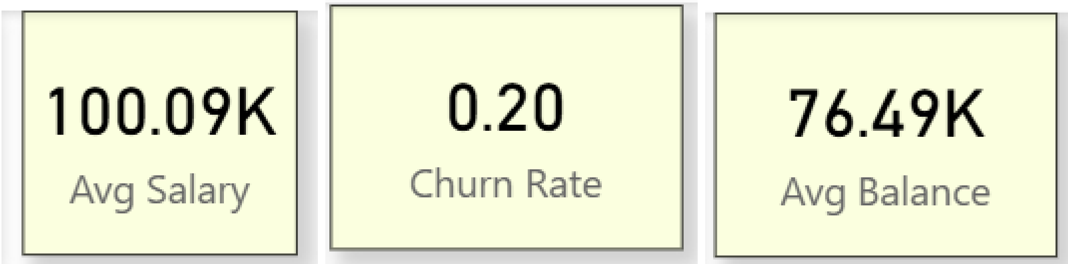




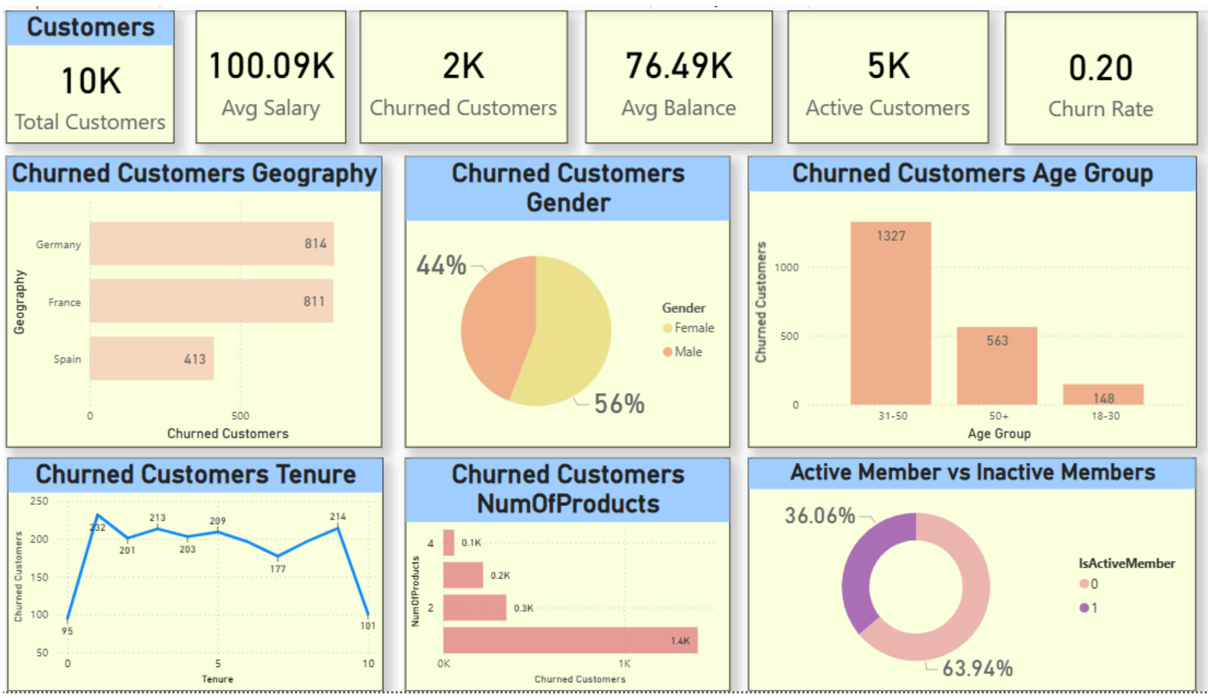




KPI cards for more understanding.



Final Dashboard-



```
#Section B: Python – Exploratory Data Analysis (Provide a description of your answers in all the cells explaining the outputs)

# 11. Check the percentage of missing data and handle accordingly.
# 12. How many rows and columns are there in the dataset?
# 13. What is the distribution of churned vs. non-churned customers?
# 14. What is the distribution of EstimatedSalary of churned and retained Customers?
# 15. How do churn rates vary by Gender, Geography, and IsActiveMember?
# 16. What is the average CreditScore, Balance, and EstimatedSalary of churned vs. retained customers?
# 17. How does Age impact churn? Plot histograms and boxplots for churned and non-churned groups.
# 18. Is there any correlation among numeric features like CreditScore, Balance, and EstimatedSalary?
# 19. What does a heatmap reveal about feature interactions with churn?
# 20. Are there outliers in Balance, CreditScore, or Age that are mostly associated with churn?
# 21. Group customers into age brackets (e.g.,18-30 as Adults, 30-50 as middle age and 50-100 as seniors.). How does churn rate
# 22. Are customers with only one product (NumOfProducts = 1) more likely to churn than those with multiple?
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/Customer-Churn-Records.csv")
```

```
# 11. Check the percentage of missing data and handle accordingly.
df.isnull().sum()
df.isnull().mean()*100
```

	0
RowNumber	0.0
CustomerId	0.0
Surname	0.0
CreditScore	0.0
Geography	0.0
Gender	0.0
Age	0.0
Tenure	0.0
Balance	0.0
NumOfProducts	0.0
HasCrCard	0.0
IsActiveMember	0.0
EstimatedSalary	0.0
Exited	0.0
Complain	0.0
Satisfaction Score	0.0
Card Type	0.0
Point Earned	0.0

dtype: float64

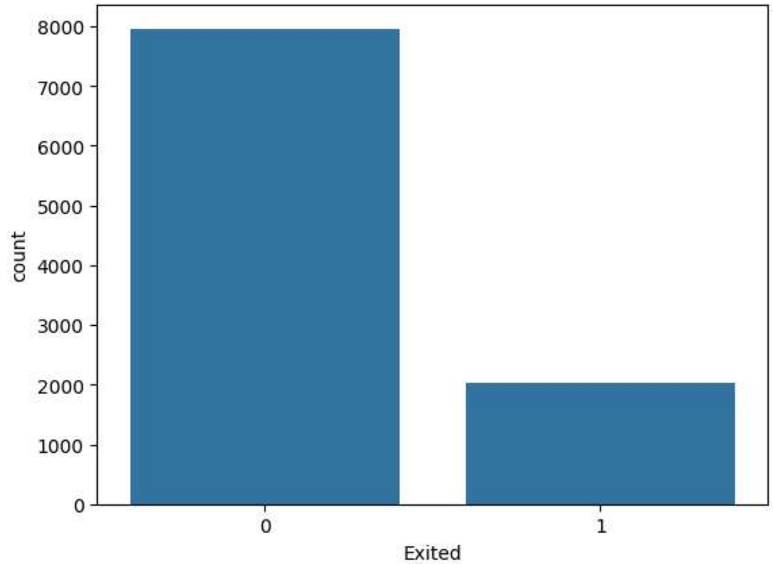
```
# 12.   How many rows and columns are there in the dataset?
df.shape
```

```
(10000, 18)
```

```
# 13.   What is the distribution of churned vs. non-churned customers?
df['Exited'].value_counts()
```

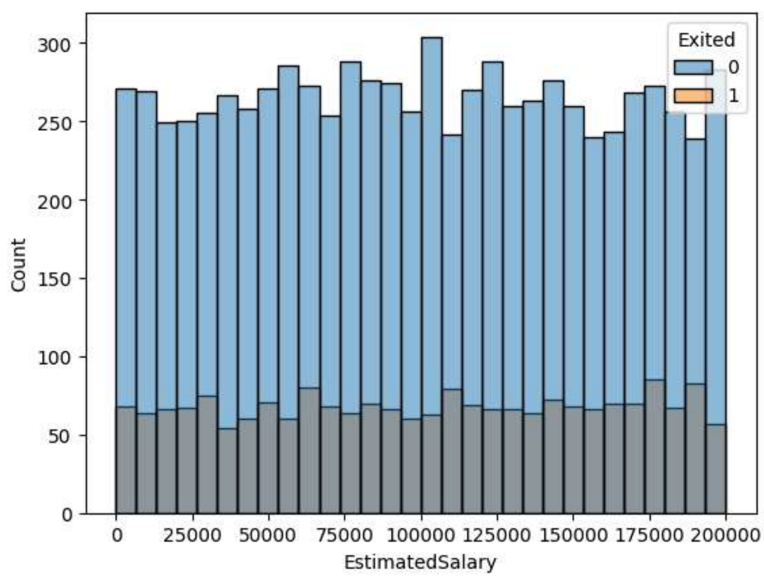
```
sns.countplot(x='Exited', data=df)
```

```
<Axes: xlabel='Exited', ylabel='count'>
```




```
# 14.   What is the distribution of EstimatedSalary of churned and retained Customers?
sns.histplot(data=df, x='EstimatedSalary', hue='Exited', bins=30)
```

```
<Axes: xlabel='EstimatedSalary', ylabel='Count'>
```




```
# 15.   How do churn rates vary by Gender, Geography, and IsActiveMember?
print(pd.crosstab(df['Gender'], df['Exited'], normalize='index') * 100)
print(pd.crosstab(df['Geography'], df['Exited'], normalize='index') * 100)
print(pd.crosstab(df['IsActiveMember'], df['Exited'], normalize='index') * 100)
```

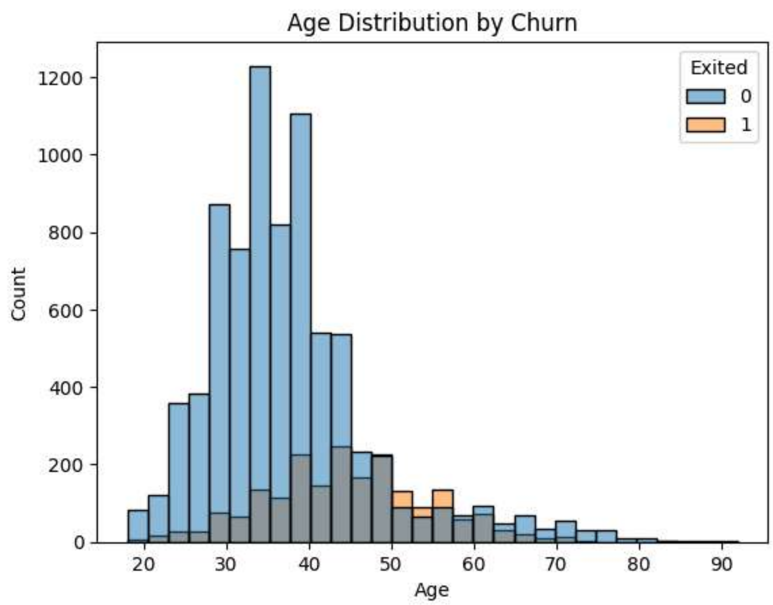
```
Exited          0          1
Gender
Female  74.928461  25.071539
Male    83.525747  16.474253
Exited             0          1
Geography
France     83.825289  16.174711
Germany    67.556796  32.443204
Spain      83.326605  16.673395
Exited                  0          1
IsActiveMember
0               73.128480  26.871520
1               85.730926  14.269074
```

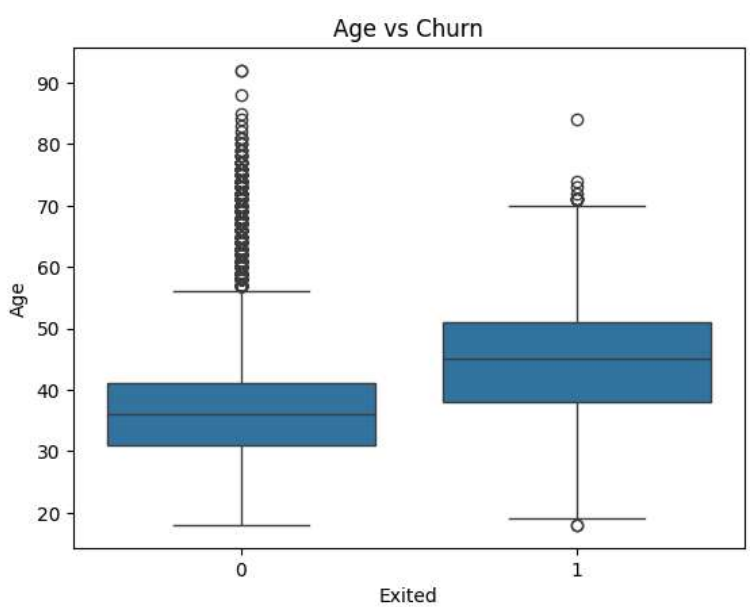
```
# 16.   What is the average CreditScore, Balance, and EstimatedSalary of churned vs. retained customers?
df.groupby('Exited')[['CreditScore','Balance','EstimatedSalary']].mean()
```

	CreditScore	Balance	EstimatedSalary
Exited			
0	651.837855	72742.750663	99726.853141
1	645.414622	91109.476006	101509.908783

```
# 17. How does Age impact churn? Plot histograms and boxplots for churned and non-churned groups.
sns.histplot(data=df, x='Age', hue='Exited', bins=30)
plt.title("Age Distribution by Churn")
plt.show()

sns.boxplot(x='Exited', y='Age', data=df)
plt.title("Age vs Churn")
plt.show()
```

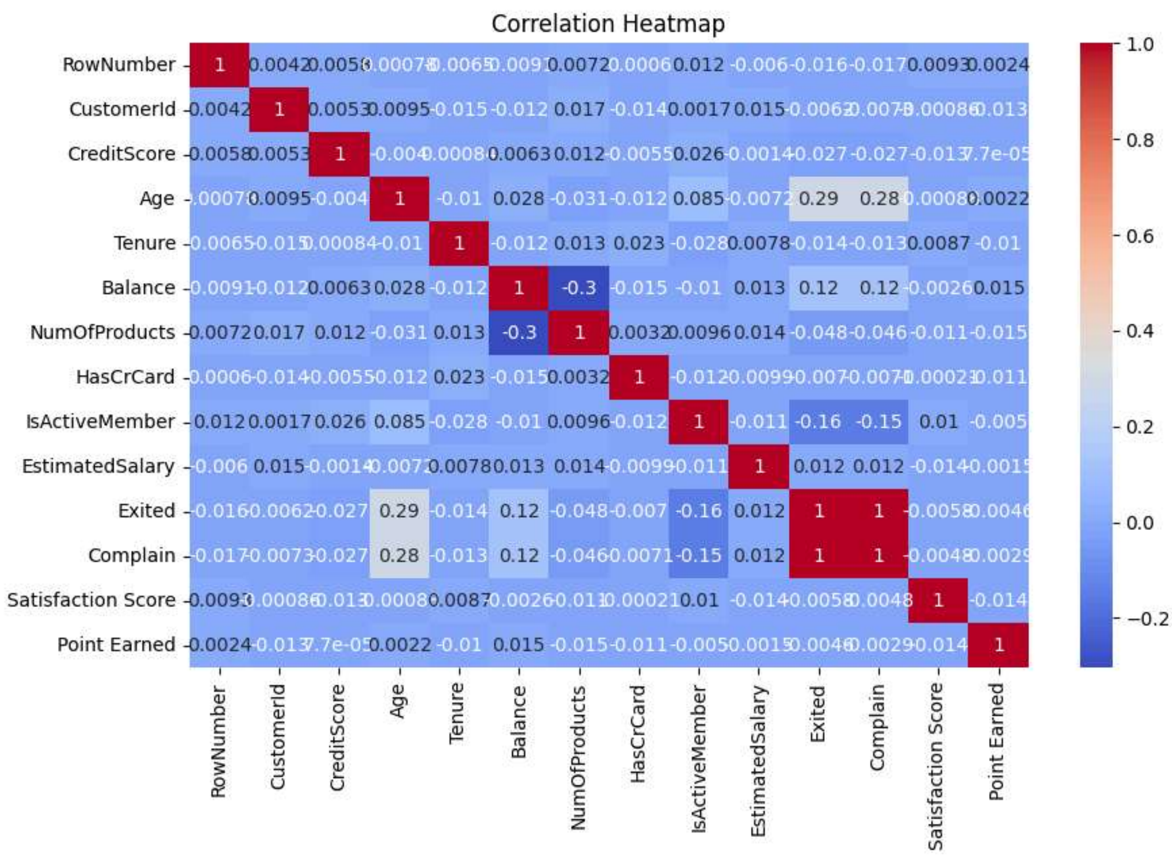





```
# 18.   Is there any correlation among numeric features like CreditScore, Balance, and EstimatedSalary?
df[['CreditScore','Balance','EstimatedSalary']].corr()
```

	CreditScore	Balance	EstimatedSalary
CreditScore	1.000000	0.006268	-0.001384
Balance	0.006268	1.000000	0.012797
EstimatedSalary	-0.001384	0.012797	1.000000

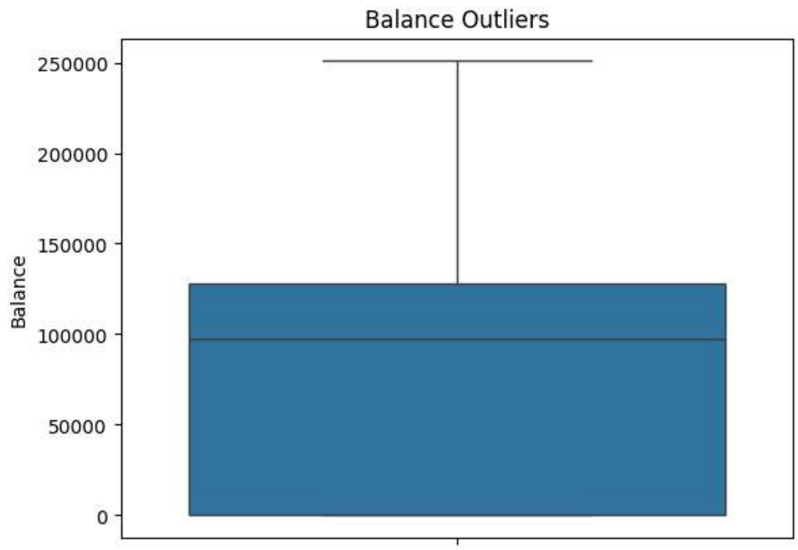
```
# 19. What does a heatmap reveal about feature interactions with churn?
plt.figure(figsize=(10,6))
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

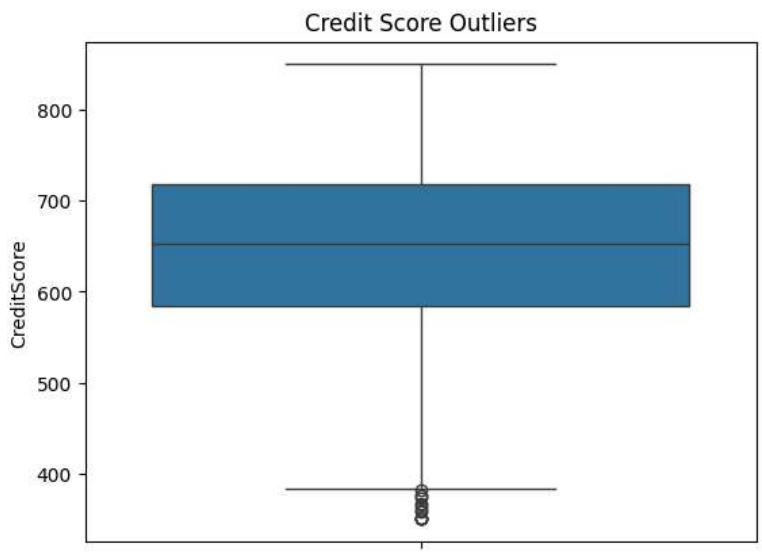


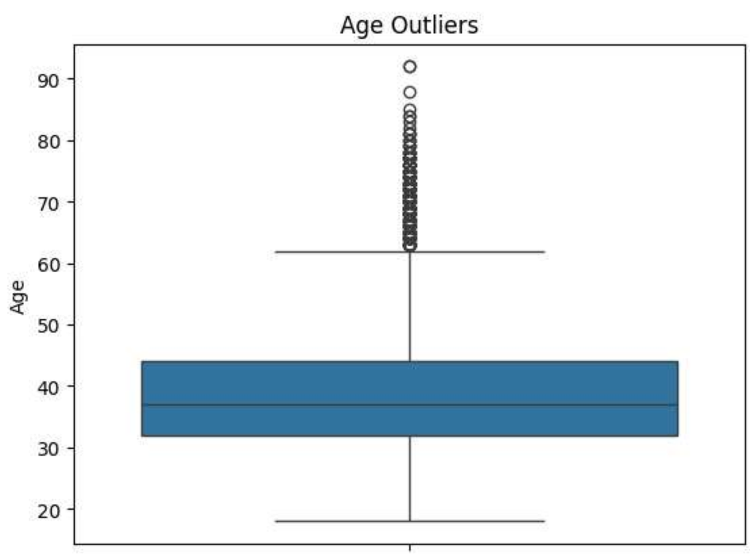

```
# 20.   Are there outliers in Balance, CreditScore, or Age that are mostly associated with churn?
sns.boxplot(y=df['Balance'])
plt.title("Balance Outliers")
plt.show()

sns.boxplot(y=df['CreditScore'])
plt.title("Credit Score Outliers")
plt.show()

sns.boxplot(y=df['Age'])
plt.title("Age Outliers")
plt.show()
```








```
# 21.   Group customers into age brackets (e.g.,18-30 as Adults, 30-50 as middle age and 50-100 as seniors.). How does churn
df['AgeGroup'] = pd.cut(df['Age'], bins=[18,30,50,100], labels=['Adult','Middle','Senior'])
pd.crosstab(df['AgeGroup'], df['Exited'], normalize='index')*100
```

Exited	0	1
AgeGroup		
Adult	92.497431	7.502569

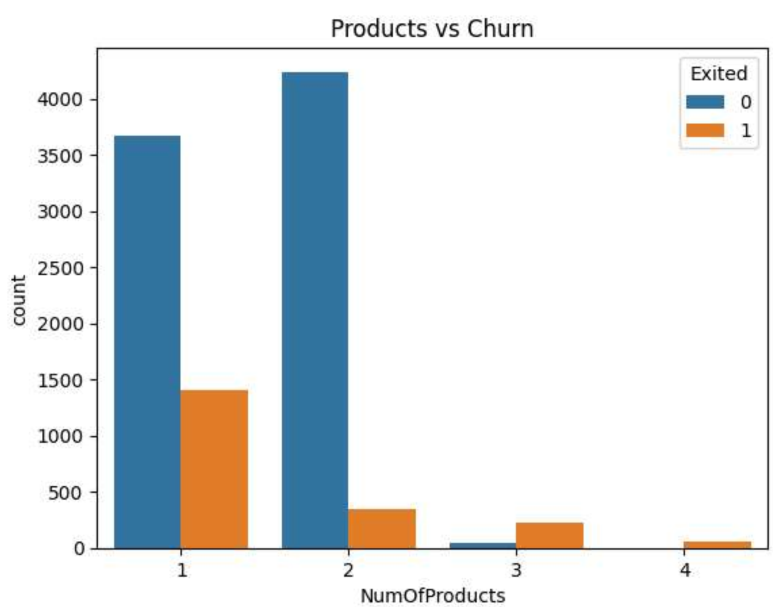
```
# 22.   Are customers with only one product (NumOfProducts = 1) more likely to churn than those with multiple?
pd.crosstab(df['NumOfProducts'], df['Exited'], normalize='index')*100
```

Exited	0	1
NumOfProducts		
1	72.285602	27.714398
2	92.396514	7.603486
3	17.293233	82.706767
4	0.000000	100.000000

```
print("Key Insights:")
print("- Inactive customers are more likely to churn")
print("- Customers with fewer products churn more")
print("- Older customers show higher churn tendency")
print("- Geography plays an important role in churn behavior")
```

```
Key Insights:
- Inactive customers are more likely to churn
- Customers with fewer products churn more
- Older customers show higher churn tendency
- Geography plays an important role in churn behavior
```

```
sns.countplot(x='NumOfProducts', hue='Exited', data=df)
plt.title("Products vs Churn")
plt.show()
```



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